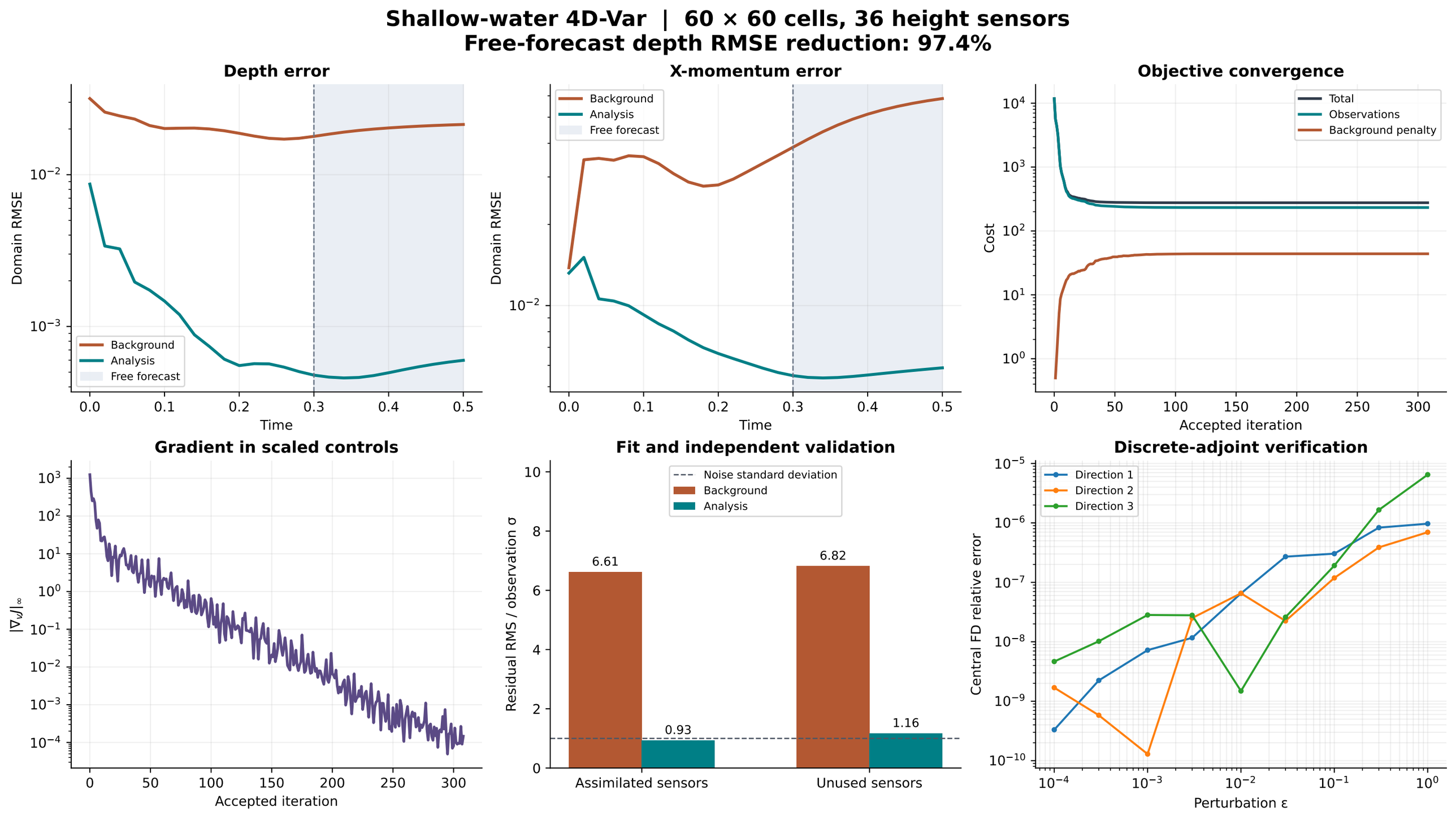
Shallow-water 4D-Var | 60 × 60 cells, 36 height sensors
Free-forecast depth RMSE reduction: 97.4%
Depth error
Domain RMSE
Time
Background
Analysis
Free forecast
X-momentum error
Domain RMSE
Time
Background
Analysis
Free forecast
Objective convergence
Cost
Accepted iteration
Total
Observations
Background penalty
Gradient in scaled controls
$|\nabla_v J|_\infty$
Accepted iteration
Fit and independent validation
Residual RMS / observation σ
Noise standard deviation
Background
Analysis
6.61
0.93
6.82
1.16
Assimilated sensors
Unused sensors
Discrete-adjoint verification
Central FD relative error
Perturbation ε
Direction 1
Direction 2
Direction 3